ERNIE Magnet Assembly Procedure

Overview

The ERNIE toolkit provides a hands-on platform for learning the principles involved in constructing a low-field MRI system. The magnet assembly process involves preparing the mechanical components, installing magnets into the 3D-printed Halbach array rings, assembling the magnet structure, and verifying the resulting magnetic field configuration.

Before the start of the assembly session, few preparation steps are completed to ensure that the components are ready for assembly.

1. Pre-Assembly Preparation

1.1 Preparation of Brass Support Rods

The brass rods used to support and align the Halbach array structure are prepared before the assembly process.

A **1m** brass rod is measured and cut into five sections, each with a length of **180 mm**. These prepared rods are used during the final assembly stage to hold the magnet rings in alignment.

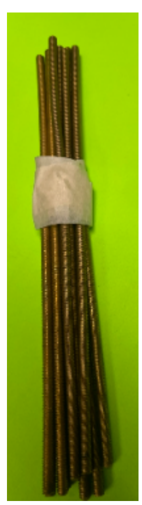


Figure 1: Brass rods cut to 180 mm length for Halbach array assembly.

1.2 Installation of Heat-Set Inserts

Due to restrictions on the use of soldering equipment in conference halls and workshop venues, the brass heat-set inserts are installed into the 3D-printed rings in a lab space, before the assembly sessions.

The inserts are installed by heating the soldering iron nozzle to the required temperature and carefully embedding each brass insert into its designated mounting hole. These inserts provide threaded attachment points for securing the ring lids.

During installation, controlled heat and gentle pressure are applied to ensure that each insert is:

- Properly aligned with the mounting hole
- Fully embedded within the printed material
- Flush with the ring surface

After installation, the rings are placed on a flat surface and allowed to cool completely before proceeding with magnet assembly.

Preparation of Ring Alignment Reference Marks

Before the assembly process, a horizontal reference line is marked on each 3D-printed ring to define the starting position for magnet placement. This reference line corresponds to the first magnet slot positioned at **0° relative to the horizontal axis** of the ring. The marked reference position provides a consistent starting point for inserting magnets and ensures correct alignment and orientation of the magnets within each Halbach array ring. This is shown in Figure 2 below.

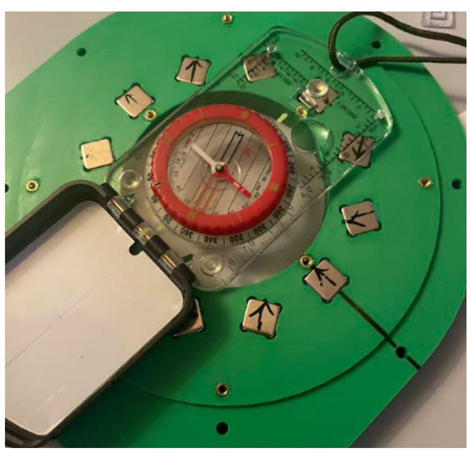


Figure 2: Ring #42 showing installed heat-set inserts and the 0° horizontal reference line for magnet alignment.

Magnet assembly

Step 1: Arrange Components

All components are removed from the storage case and systematically arranged on a flat working surface.This allows participants to inspect the components, familiarize themselves with the assembly sequence, and prepare the workspace before beginning the build process.

Step 2: Arrange Neodymium Magnets

The neodymium magnets are removed from their storage boxes and arranged vertically in an organized manner. This arrangement facilitates efficient polarity identification and ensures that magnets can be handled safely during the assembly process.

Step 3: Identify the Reference Magnet Polarity

From the batch of neodymium magnets, **four magnets are selected to be prepared as reference magnets** for establishing the polarity orientation of the remaining magnets. One of the selected magnets is first used to determine its magnetic pole direction. A compass is used to identify the direction of the Earth's magnetic field and is then brought close to the selected magnet without allowing physical contact. The magnet face that attracts the north-seeking end of the compass is identified as the **south pole**, while the opposite face corresponds to the **north pole**. The north pole direction of the magnet is marked using a permanent marker, transforming it into a reference magnet. A small cardboard separator is attached to the opposite face using masking tape to provide physical separation and prevent unintended magnetic interactions during the polarity identification of the remaining magnets.

Step 4 – Polarity Identification and Marking of Remaining Magnets

A row of magnets is carefully brought into proximity with the reference magnet while maintaining controlled handling to prevent sudden magnetic attraction forces, which could cause the magnets to collide and potentially crack. This process is illustrated in Figure 3 below. The face of each magnet within the row that is attracted to the south pole of the reference magnet is identified as the **north pole** and marked accordingly. An arrow is drawn on each magnet to indicate the north pole direction. This procedure is repeated symmetrically for all remaining rows of magnets until the polarity of every magnet has been identified and clearly marked using permanent markers.

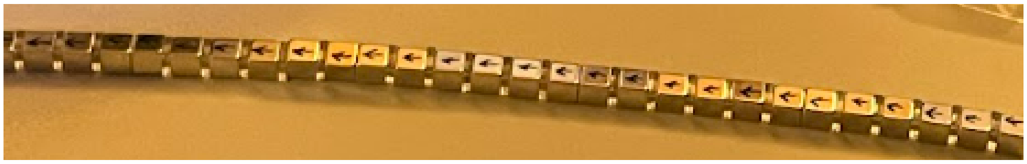


Figure 3: Row of magnets arranged for polarity identification and marking.

Step 5 – Placing Magnets into Ring Slots

Before installation, each neodymium magnet is inspected for visible defects, such as cracks or dents, that may affect its magnetic performance. The magnetic field strength of individual magnets can be verified using a Gauss meter by measuring the surface magnetic field, with **N48** neodymium magnets expected to produce a field strength greater than **1 T** near the magnet surface. The magnets are then inserted into their corresponding ring slots according to their marked polarities, starting from the slot positioned at 0° relative to the horizontal axis. During insertion, the arrow marking on each magnet is aligned with the circular indicator on the slot to ensure correct orientation. Figure 4 shows the magnet alignment in Ring #42.

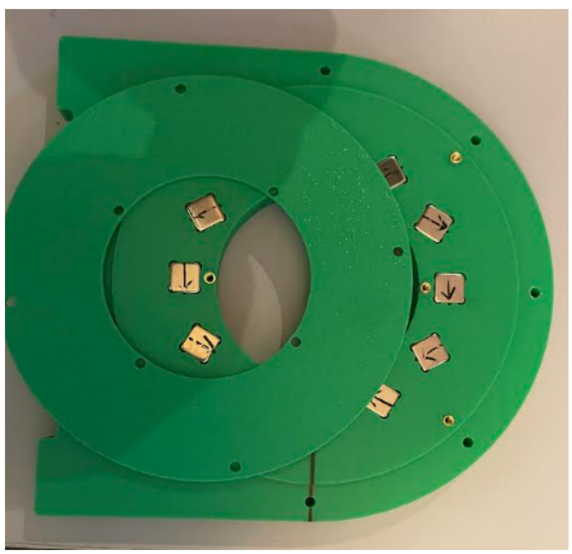


Figure 4: Showing alignment of Neodymium magnets in Ring #42.

Step 6 – Verification of Magnet Orientation

After completing magnet placement in a ring, a second participant verifies that the magnets are inserted in the correct order and orientation. A compass is then used to confirm the Halbach array configuration (Figure 5). At the center of the ring, the compass should indicate a consistent magnetic field direction, while movement around the exterior of the ring causes the compass needle to rotate, confirming the expected internal and external field behavior of the Halbach array. This verification is repeated for all rings.

Figure 5: Verification of magnet orientation within the ring using a compass.

Step 7 – Securing Ring Lids

The corresponding lids are attached to each ring using M3 brass screws and screwdrivers. The brass screws are tightened securely to ensure that the magnets remain fixed within their slots during handling and subsequent assembly steps.

Figure 6: Rings containing magnets with lids securely attached.

Step 8 – Assembly of the Halbach Array

The Halbach array assembly begins by preparing the eight brass rods. A dome-shaped nut is first placed onto the end of each brass rod, and the rods are positioned vertically. Nylon washers are then added on top of the dome-shaped nuts to support the first ring.

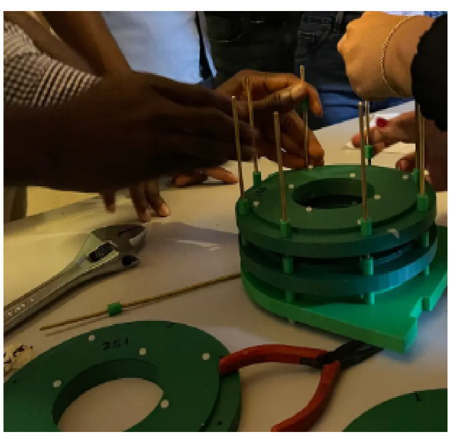
The rings are assembled onto the brass rods in sequence. As each ring is placed, the **lid side is kept facing upwards**, and care is taken to ensure that the horizontal reference line marked on each ring remains aligned in the same direction throughout the assembly. This alignment maintains the correct relative orientation of the magnet rings within the Halbach array.

Ring #42 is first placed onto the eight brass rods. **An individual ring spacer is then placed onto each of the eight brass rods above the ring** before installing the next ring. The remaining rings and spacers are assembled in the following sequence:

Ring #42 → 8 Spacers → Ring #251 → 8 Spacers → Ring #288 → 8 Spacers → Ring #321 → 8 Spacers → Ring #288 → 8 Spacers → Ring #251 → 8 Spacers → Ring #42

After the final Ring #42 is installed, nylon washers are added onto the eight brass rods, and the assembly is secured using brass nuts tightened with spanners.

During the tightening process, a vernier caliper is used to measure the distance between adjacent rings and ensure that the spacing is maintained at **12 mm**. If adjustments are required, the position of the rings is slightly modified while tightening the brass nuts until the correct spacing is achieved.



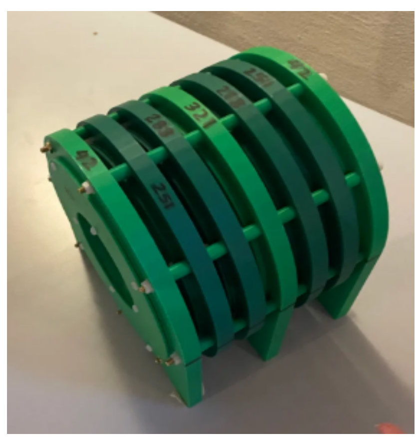
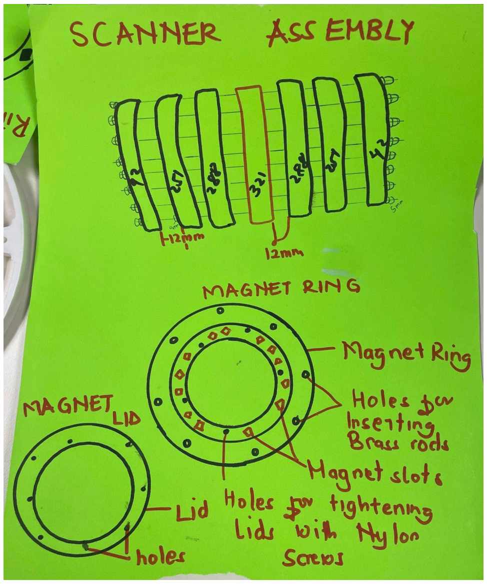
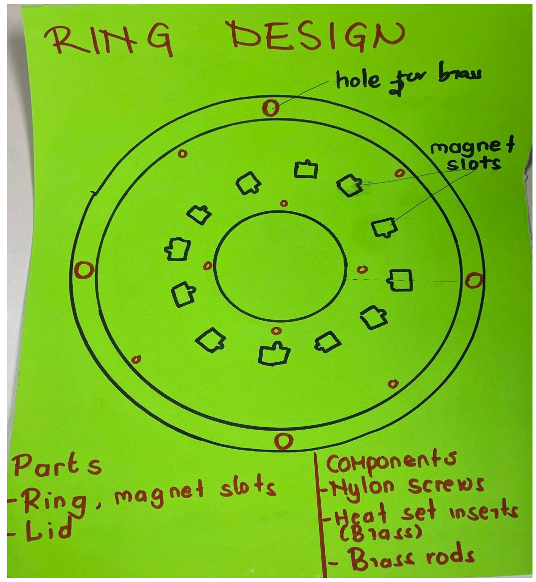


Figure 7: Halbach array ring assembly process and completed magnet assembly.

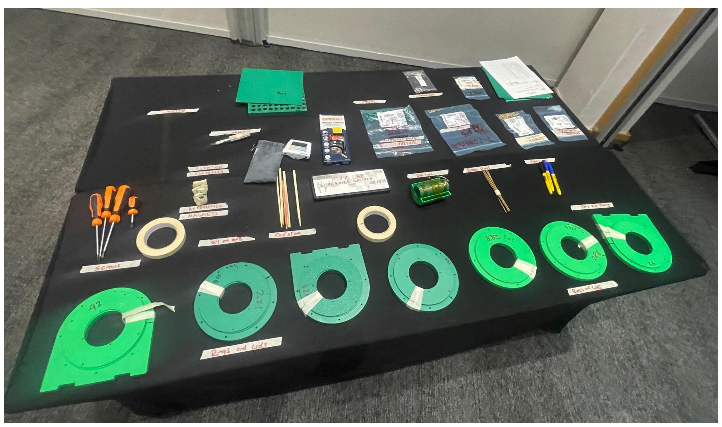
Supplementary Images

Visual assembly guides illustrating the scanner configuration and ring arrangement.

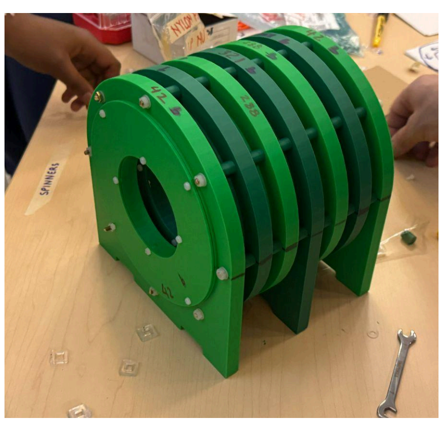




Pre-assembly arrangement of scanner components and required tools.



Assembled Scanner



Magnets in disassembly tray.